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RIP Version 2

Status of this Memo

This document specifies an Internet standards track protocol for the Internet community, and requests discussion and suggestions for improvements. Please refer to the current edition of the "Internet Official Protocol Standards" (STD 1) for the standardization state and status of this protocol. Distribution of this memo is unlimited.

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Abstract

This document specifies an extension of the Routing Information Protocol (RIP), as defined in [1], to expand the amount of useful information carried in RIP messages and to add a measure of security.

A companion document will define the SNMP MIB objects for RIP-2 [2]. An additional document will define cryptographic security improvements for RIP-2 [3].

Acknowledgements

I would like to thank the IETF RIP Working Group for their help in improving the RIP-2 protocol. Much of the text for the background discussions about distance vector protocols and some of the descriptions of the operation of RIP were taken from "Routing Information Protocol" by C. Hedrick [1]. Some of the final editing on the document was done by Scott Bradner.

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1. Justification

With the advent of OSPF and IS-IS, there are those who believe that RIP is obsolete. While it is true that the newer IGP routing protocols are far superior to RIP, RIP does have some advantages. Primarily, in a small network, RIP has very little overhead in terms of bandwidth used and configuration and management time. RIP is also very easy to implement, especially in relation to the newer IGPs.

Additionally, there are many, many more RIP implementations in the field than OSPF and IS-IS combined. It is likely to remain that way for some years yet.

Given that RIP will be useful in many environments for some period of time, it is reasonable to increase RIP's usefulness. This is especially true since the gain is far greater than the expense of the change.

2. Current RIP

The current RIP-1 message contains the minimal amount of information necessary for routers to route messages through a network. It also contains a large amount of unused space, owing to its origins.

The current RIP-1 protocol does not consider autonomous systems and IGP/EGP interactions, subnetting [11], and authentication since implementations of these postdate RIP-1. The lack of subnet masks is a particularly serious problem for routers since they need a subnet mask to know how to determine a route. If a RIP-1 route is a network route (all non-network bits 0), the subnet mask equals the network mask. However, if some of the non-network bits are set, the router cannot determine the subnet mask. Worse still, the router cannot determine if the RIP-1 route is a subnet route or a host route. Currently, some routers simply choose the subnet mask of the interface over which the route was learned and determine the route type from that.

3. Basic Protocol

3.1 Introduction

RIP is a routing protocol based on the **Bellman-Ford** (or distance vector) algorithm. This algorithm has been used for routing computations in computer networks since the early days of the ARPANET. The particular packet formats and protocol described here are based on the program "routed," which is included with the Berkeley distribution of Unix.

In an international network, such as the Internet, it is very unlikely that a single routing protocol will be used for the entire network. Rather, the network will be organized as a collection of Autonomous Systems (AS), each of which will, in general, be administered by a single entity. Each AS will have its own routing technology, which may differ among AS's. The routing protocol used within an AS is referred to as an Interior Gateway Protocol (IGP). A separate protocol, called an Exterior Gateway Protocol (EGP), is used to transfer routing information among the AS's. RIP was designed to work as an IGP in moderate-size AS's. It is not intended for use in more complex environments. For information on the context into which RIP-1 is expected to fit, see Braden and Postel [6].

RIP uses one of a class of routing algorithms known as Distance Vector algorithms. The earliest description of this class of algorithms known to the author is in Ford and Fulkerson [8]. Because of this, they are sometimes known as Ford-Fulkerson algorithms. The term Bellman-Ford is also used, and derives from the fact that the formulation is based on Bellman's equation [4]. The presentation in this document is closely based on [5]. This document contains a protocol specification. For an introduction to the mathematics of routing algorithms, see [1]. The basic algorithms used by this protocol were used in computer routing as early as 1969 in the ARPANET. However, the specific ancestry of this protocol is within the Xerox network protocols. The PUP protocols [7] used the Gateway Information Protocol to exchange routing information. A somewhat updated version of this protocol was adopted for the Xerox Network Systems (XNS) architecture, with the name Routing Information Protocol [9]. Berkeley's routed is largely the same as the Routing Information Protocol, with XNS addresses replaced by a more general address format capable of handling IPv4 and other types of address, and with routing updates limited to one every 30 seconds. Because of this similarity, the term Routing Information Protocol (or just RIP) is used to refer to both the XNS protocol and the protocol used by routed.

RIP is intended for use within the IP-based Internet. The Internet is organized into a number of networks connected by special purpose gateways known as routers. The networks may be either point-to-point links or more complex networks such as Ethernet or token ring. Hosts and routers are presented with IP datagrams addressed to some host. Routing is the method by which the host or router decides where to send the datagram. It may be able to send the datagram directly to the destination, if that destination is on one of the networks that are directly connected to the host or router. However, the interesting case is when the destination is not directly reachable.

In this case, the host or router attempts to send the datagram to a router that is nearer the destination. The goal of a routing protocol is very simple: It is to supply the information that is needed to do routing.

3.2 Limitations of the Protocol

This protocol does not solve every possible routing problem. As mentioned above, it is primary intended for use as an IGP in networks of moderate size. In addition, the following specific limitations are be mentioned:

- The protocol is limited to networks whose longest path (the network's diameter) is 15 hops. The designers believe that the basic protocol design is inappropriate for larger networks. Note that this statement of the limit assumes that a cost of 1 is used for each network. This is the way RIP is normally configured. If the system administrator chooses to use larger costs, the upper bound of 15 can easily become a problem.
- The protocol depends upon "counting to infinity" to resolve certain unusual situations. (This will be explained in the next section.) If the system of networks has several hundred networks, and a routing loop was formed involving all of them, the resolution of the loop would require either much time (if the frequency of routing updates were limited) or bandwidth (if updates were sent whenever changes were detected). Such a loop would consume a large amount of network bandwidth before the loop was corrected. We believe that in realistic cases, this will not be a problem except on slow lines. Even then, the problem will be fairly unusual, since various precautions are taken that should prevent these problems in most cases.
- This protocol uses fixed "metrics" to compare alternative routes. It is not appropriate for situations where routes need to be chosen based on real-time parameters such a measured delay, reliability, or load. The obvious extensions to allow metrics of this type are likely to introduce instabilities of a sort that the protocol is not designed to handle.

3.3. Organization of this document

The main body of this document is organized into two parts, which occupy the next two sections:

A conceptual development and justification of distance vector algorithms in general.

The actual protocol description.

Each of these two sections can largely stand on its own. Section 3.4 attempts to give an informal presentation of the mathematical underpinnings of the algorithm. Note that the presentation follows a "spiral" method. An initial, fairly simple algorithm is described. Then refinements are added to it in successive sections. Section 3.5 is the actual protocol description. Except where specific references are made to section 3.4, it should be possible to implement RIP entirely from the specifications given in section 3.5.

3.4 Distance Vector Algorithms

Routing is the task of finding a path from a sender to a desired destination. In the IP "Internet model" this reduces primarily to a matter of finding a series of routers between the source and destination networks. As long as a message or datagram remains on a single network or subnet, any forwarding problems are the responsibility of technology that is specific to the network. For example, Ethernet and the ARPANET each define a way in which any sender can talk to any specified destination within that one network. IP routing comes in primarily when messages must go from a sender on one network to a destination on a different one. In that case, the message must pass through one or more routers connecting the networks. If the networks are not adjacent, the message may pass through several intervening networks, and the routers connecting them. Once the message gets to a router that is on the same network as the destination, that network's own technology is used to get to the destination.

Throughout this section, the term "network" is used generically to cover a single broadcast network (e.g., an Ethernet), a point to point line, or the ARPANET. The critical point is that a network is treated as a single entity by IP. Either no forwarding decision is necessary (as with a point to point line), or that forwarding is done in a manner that is transparent to IP, allowing IP to treat the entire network as a single fully-connected system (as with an Ethernet or the ARPANET). Note that the term "network" is used in a somewhat different way in discussions of IP addressing. We are using the term "network" here to refer to subnets in cases where subnet

addressing is in use.

A number of different approaches for finding routes between networks are possible. One useful way of categorizing these approaches is on the basis of the type of information the routers need to exchange in order to be able to find routes. Distance vector algorithms are based on the exchange of only a small amount of information. Each entity (router or host) that participates in the routing protocol is assumed to keep information about all of the destinations within the system. Generally, information about all entities connected to one network is summarized by a single entry, which describes the route to all destinations on that network. This summarization is possible because as far as IP is concerned, routing within a network is invisible. Each entry in this routing database includes the next router to which datagrams destined for the entity should be sent. In addition, it includes a "metric" measuring the total distance to the entity. Distance is a somewhat generalized concept, which may cover the time delay in getting messages to the entity, the dollar cost of sending messages to it, etc. Distance vector algorithms get their name from the fact that it is possible to compute optimal routes when the only information exchanged is the list of these distances. Furthermore, information is only exchanged among entities that are adjacent, that is, entities that share a common network.

Although routing is most commonly based on information about networks, it is sometimes necessary to keep track of the routes to individual hosts. The RIP protocol makes no formal distinction between networks and hosts. It simply describes exchange of information about destinations, which may be either networks or hosts. (Note however, that it is possible for an implementor to choose not to support host routes. See section 3.2.) In fact, the mathematical developments are most conveniently thought of in terms of routes from one host or router to another. When discussing the algorithm in abstract terms, it is best to think of a routing entry for a network as an abbreviation for routing entries for all of the entities connected to that network. This sort of abbreviation makes sense only because we think of networks as having no internal structure that is visible at the IP level. Thus, we will generally assign the same distance to every entity in a given network.

We said above that each entity keeps a routing database with one entry for every possible destination in the system. An actual implementation is likely to need to keep the following information about each destination:

- address: in IP implementations of these algorithms, this will be the IP address of the host or network.
- router: the first router along the route to the destination.
- interface: the physical network which must be used to reach the first router.
- metric: a number, indicating the distance to the destination.
- timer: the amount of time since the entry was last updated.

In addition, various flags and other internal information will probably be included. This database is initialized with a description of the entities that are directly connected to the system. It is updated according to information received in messages from neighboring routers.

The most important information exchanged by the hosts and routers is carried in update messages. Each entity that participates in the routing scheme sends update messages that describe the routing database as it currently exists in that entity. It is possible to maintain optimal routes for the entire system by using only information obtained from neighboring entities. The algorithm used for that will be described in the next section.

As we mentioned above, the purpose of routing is to find a way to get datagrams to their ultimate destinations. Distance vector algorithms are based on a table in each router listing the best route to every destination in the system. Of course, in order to define which route is best, we have to have some way of measuring goodness. This is referred to as the "metric".

In simple networks, it is common to use a metric that simply counts **how many routers a message must go through**. In more complex networks, a metric is chosen to represent the total amount of delay that the message suffers, the cost of sending it, or some other quantity which may be minimized. The main requirement is that it must be possible to represent the metric as a sum of "costs" for individual hops.

Formally, if it is possible to get from entity *i* to entity *j* directly (i.e., without passing through another router between), then a cost, $d(i,j)$, is associated with the hop between *i* and *j*. In the normal case where all entities on a given network are considered to be the same, $d(i,j)$ is the same for all destinations on a given network, and represents the cost of using that network. To get the metric of a complete route, one just adds up the costs of the individual hops

that make up the route. For the purposes of this memo, we assume that the costs are positive integers.

Let $D(i,j)$ represent the metric of the best route from entity i to entity j . It should be defined for every pair of entities. $d(i,j)$ represents the costs of the individual steps. Formally, let $d(i,j)$ represent the cost of going directly from entity i to entity j . It is infinite if i and j are not immediate neighbors. (Note that $d(i,i)$ is infinite. That is, we don't consider there to be a direct connection from a node to itself.) Since costs are additive, it is easy to show that the best metric must be described by

$$\begin{aligned} D(i,i) &= 0, & \text{all } i \\ D(i,j) &= \min_k [d(i,k) + D(k,j)], & \text{otherwise} \end{aligned}$$

and that the best routes start by going from i to those neighbors k for which $d(i,k) + D(k,j)$ has the minimum value. (These things can be shown by induction on the number of steps in the routes.) Note that we can limit the second equation to k 's that are immediate neighbors of i . For the others, $d(i,k)$ is infinite, so the term involving them can never be the minimum.

It turns out that one can compute the metric by a simple algorithm based on this. Entity i gets its neighbors k to send it their estimates of their distances to the destination j . When i gets the estimates from k , it adds $d(i,k)$ to each of the numbers. This is simply the cost of traversing the network between i and k . Now and then i compares the values from all of its neighbors and picks the smallest.

A proof is given in [2] that this algorithm will converge to the correct estimates of $D(i,j)$ in finite time in the absence of topology changes. The authors make very few assumptions about the order in which the entities send each other their information, or when the min is recomputed. Basically, entities just can't stop sending updates or recomputing metrics, and the networks can't delay messages forever. (Crash of a routing entity is a topology change.) Also, their proof does not make any assumptions about the initial estimates of $D(i,j)$, except that they must be non-negative. The fact that these fairly weak assumptions are good enough is important. Because we don't have to make assumptions about when updates are sent, it is safe to run the algorithm asynchronously. That is, each entity can send updates according to its own clock. Updates can be dropped by the network, as long as they don't all get dropped. Because we don't have to make assumptions about the starting condition, the algorithm can handle changes. When the system changes, the routing algorithm starts moving to a new equilibrium, using the old one as its starting point. It is important that the algorithm will converge in finite

time no matter what the starting point. Otherwise certain kinds of changes might lead to non-convergent behavior.

The statement of the algorithm given above (and the proof) assumes that each entity keeps copies of the estimates that come from each of its neighbors, and now and then does a min over all of the neighbors. In fact real implementations don't necessarily do that. They simply remember the best metric seen so far, and the identity of the neighbor that sent it. They replace this information whenever they see a better (smaller) metric. This allows them to compute the minimum incrementally, without having to store data from all of the neighbors.

There is one other difference between the algorithm as described in texts and those used in real protocols such as RIP: the description above would have each entity include an entry for itself, showing a distance of zero. In fact this is not generally done. Recall that all entities on a network are normally summarized by a single entry for the network. Consider the situation of a host or router G that is connected to network A. C represents the cost of using network A (usually a metric of one). (Recall that we are assuming that the internal structure of a network is not visible to IP, and thus the cost of going between any two entities on it is the same.) In principle, G should get a message from every other entity H on network A, showing a cost of 0 to get from that entity to itself. G would then compute $C + 0$ as the distance to H. Rather than having G look at all of these identical messages, it simply starts out by making an entry for network A in its table, and assigning it a metric of C. This entry for network A should be thought of as summarizing the entries for all other entities on network A. The only entity on A that can't be summarized by that common entry is G itself, since the cost of going from G to G is 0, not C. But since we never need those 0 entries, we can safely get along with just the single entry for network A. Note one other implication of this strategy: because we don't need to use the 0 entries for anything, hosts that do not function as routers don't need to send any update messages. Clearly hosts that don't function as routers (i.e., hosts that are connected to only one network) can have no useful information to contribute other than their own entry $D(i,i) = 0$. As they have only the one interface, it is easy to see that a route to any other network through them will simply go in that interface and then come right back out it. Thus the cost of such a route will be greater than the best cost by at least C. Since we don't need the 0 entries, non-routers need not participate in the routing protocol at all.

Let us summarize what a host or router G does. For each destination in the system, G will keep a current estimate of the metric for that destination (i.e., the total cost of getting to it) and the identity

of the neighboring router on whose data that metric is based. If the destination is on a network that is directly connected to G, then G simply uses an entry that shows the cost of using the network, and the fact that no router is needed to get to the destination. It is easy to show that once the computation has converged to the correct metrics, the neighbor that is recorded by this technique is in fact the first router on the path to the destination. (If there are several equally good paths, it is the first router on one of them.) This combination of destination, metric, and router is typically referred to as a route to the destination with that metric, using that router.

4.ne The method so far only has a way to lower the metric, as the existing metric is kept until a smaller one shows up. It is possible that the initial estimate might be too low. Thus, there must be a way to increase the metric. It turns out to be sufficient to use the following rule: suppose the current route to a destination has metric D and uses router G. If a new set of information arrived from some source other than G, only update the route if the new metric is better than D. But if a new set of information arrives from G itself, always update D to the new value. It is easy to show that with this rule, the incremental update process produces the same routes as a calculation that remembers the latest information from all the neighbors and does an explicit minimum. (Note that the discussion so far assumes that the network configuration is static. It does not allow for the possibility that a system might fail.)

To summarize, here is the basic distance vector algorithm as it has been developed so far. (Note that this is not a statement of the RIP protocol. There are several refinements still to be added.) The following procedure is carried out by every entity that participates in the routing protocol. This must include all of the routers in the system. Hosts that are not routers may participate as well.

- Keep a table with an entry for every possible destination in the system. The entry contains the distance D to the destination, and the first router G on the route to that network. Conceptually, there should be an entry for the entity itself, with metric 0, but this is not actually included.
- Periodically, send a routing update to every neighbor. The update is a set of messages that contain all of the information from the routing table. It contains an entry for each destination, with the distance shown to that destination.
- When a routing update arrives from a neighbor G', add the cost associated with the network that is shared with G'. (This should be the network over which the update arrived.) Call the resulting

distance D' . Compare the resulting distances with the current routing table entries. If the new distance D' for N is smaller than the existing value D , adopt the new route. That is, change the table entry for N to have metric D' and router G' . If G' is the router from which the existing route came, i.e., $G' = G$, then use the new metric even if it is larger than the old one.

3.4.1 Dealing with changes in topology

The discussion above assumes that the topology of the network is fixed. In practice, routers and lines often fail and come back up. To handle this possibility, we need to modify the algorithm slightly.

The theoretical version of the algorithm involved a minimum over all immediate neighbors. If the topology changes, the set of neighbors changes. Therefore, the next time the calculation is done, the change will be reflected. However, as mentioned above, actual implementations use an incremental version of the minimization. Only the best route to any given destination is remembered. If the router involved in that route should crash, or the network connection to it break, the calculation might never reflect the change. The algorithm as shown so far depends upon a router notifying its neighbors if its metrics change. If the router crashes, then it has no way of notifying neighbors of a change.

In order to handle problems of this kind, distance vector protocols must make some provision for timing out routes. The details depend upon the specific protocol. As an example, in RIP every router that participates in routing sends an update message to all its neighbors once every 30 seconds. Suppose the current route for network N uses router G . If we don't hear from G for 180 seconds, we can assume that either the router has crashed or the network connecting us to it has become unusable. Thus, we mark the route as invalid. When we hear from another neighbor that has a valid route to N , the valid route will replace the invalid one. Note that we wait for 180 seconds before timing out a route even though we expect to hear from each neighbor every 30 seconds. Unfortunately, messages are occasionally lost by networks. Thus, it is probably not a good idea to invalidate a route based on a single missed message.

As we will see below, it is useful to have a way to notify neighbors that there currently isn't a valid route to some network. RIP, along with several other protocols of this class, does this through a normal update message, by marking that network as unreachable. A specific metric value is chosen to indicate an unreachable destination; that metric value is larger than the largest valid metric that we expect to see. In the existing implementation of RIP, 16 is used. This value is normally referred to as "infinity", since

it is larger than the largest valid metric. 16 may look like a surprisingly small number. It is chosen to be this small for reasons that we will see shortly. In most implementations, the same convention is used internally to flag a route as invalid.

3.4.2 Preventing instability

The algorithm as presented up to this point will always allow a host or router to calculate a correct routing table. However, that is still not quite enough to make it useful in practice. The proofs referred to above only show that the routing tables will converge to the correct values in finite time. They do not guarantee that this time will be small enough to be useful, nor do they say what will happen to the metrics for networks that become inaccessible.

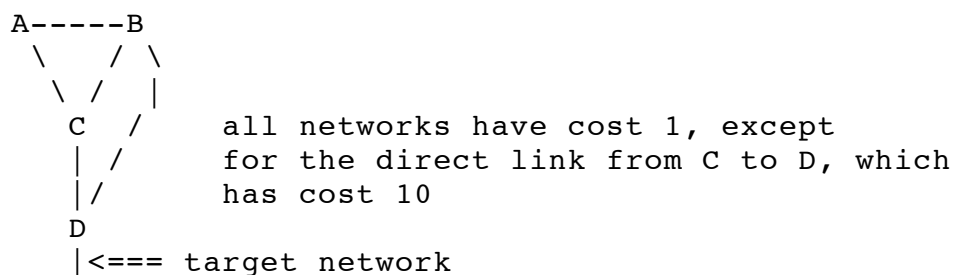
It is easy enough to extend the mathematics to handle routes becoming inaccessible. The convention suggested above will do that. We choose a large metric value to represent "infinity". This value must be large enough that no real metric would ever get that large. For the purposes of this example, we will use the value 16. Suppose a network becomes inaccessible. All of the immediately neighboring routers time out and set the metric for that network to 16. For purposes of analysis, we can assume that all the neighboring routers have gotten a new piece of hardware that connects them directly to the vanished network, with a cost of 16. Since that is the only connection to the vanished network, all the other routers in the system will converge to new routes that go through one of those routers. It is easy to see that once convergence has happened, all the routers will have metrics of at least 16 for the vanished network. Routers one hop away from the original neighbors would end up with metrics of at least 17; routers two hops away would end up with at least 18, etc. As these metrics are larger than the maximum metric value, they are all set to 16. It is obvious that the system will now converge to a metric of 16 for the vanished network at all routers.

Unfortunately, the question of how long convergence will take is not amenable to quite so simple an answer. Before going any further, it will be useful to look at an example (taken from [2]). Note that what we are about to show will not happen with a correct implementation of RIP. We are trying to show why certain features are needed. In the following example the letters correspond to routers, and the lines to networks.

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Each router will have a table showing a route to each network.

However, for purposes of this illustration, we show only the routes from each router to the network marked at the bottom of the diagram.

```

D:  directly connected, metric 1
B:  route via D, metric 2
C:  route via B, metric 3
A:  route via B, metric 3

```

Now suppose that the link from B to D fails. The routes should now adjust to use the link from C to D. Unfortunately, it will take a while for this to happen. The routing changes start when B notices that the route to D is no longer usable. For simplicity, the chart below assumes that all routers send updates at the same time. The chart shows the metric for the target network, as it appears in the routing table at each router.

time ----->

```

D: dir, 1   dir, 1   dir, 1   dir, 1   ...   dir, 1   dir, 1
B: unreach C,   4   C,   5   C,   6           C,  11   C,  12
C: B,   3   A,   4   A,   5   A,   6           A,  11   D,  11
A: B,   3   C,   4   C,   5   C,   6           C,  11   C,  12

```

```

dir = directly connected
unreach = unreachable

```

Here's the problem: B is able to get rid of its failed route using a timeout mechanism, but vestiges of that route persist in the system for a long time. Initially, A and C still think they can get to D via B. So, they keep sending updates listing metrics of 3. In the next iteration, B will then claim that it can get to D via either A or C. Of course, it can't. The routes being claimed by A and C are now gone, but they have no way of knowing that yet. And even when they discover that their routes via B have gone away, they each think there is a route available via the other. Eventually the system converges, as all the mathematics claims it must. But it can take some time to do so. The worst case is when a network becomes

completely inaccessible from some part of the system. In that case, the metrics may increase slowly in a pattern like the one above until they finally reach infinity. For this reason, the problem is called "counting to infinity".

You should now see why "infinity" is chosen to be as small as possible. If a network becomes completely inaccessible, we want counting to infinity to be stopped as soon as possible. Infinity must be large enough that no real route is that big. But it shouldn't be any bigger than required. Thus the choice of infinity is a tradeoff between network size and speed of convergence in case counting to infinity happens. The designers of RIP believed that the protocol was unlikely to be practical for networks with a diameter larger than 15.

There are several things that can be done to prevent problems like this. The ones used by RIP are called "split horizon with poisoned reverse", and "triggered updates".

3.4.3 Split horizon

Note that some of the problem above is caused by the fact that A and C are engaged in a pattern of mutual deception. Each claims to be able to get to D via the other. This can be prevented by being a bit more careful about where information is sent. In particular, it is never useful to claim reachability for a destination network to the neighbor(s) from which the route was learned. "Split horizon" is a scheme for **avoiding problems caused by including routes in updates sent to the router from which they were learned.** The "simple split horizon" scheme omits routes learned from one neighbor in updates sent to that neighbor. "Split horizon with poisoned reverse" includes such routes in updates, but sets their metrics to infinity.

If A thinks it can get to D via C, its messages to C should indicate that D is unreachable. If the route through C is real, then C either has a direct connection to D, or a connection through some other router. C's route can't possibly go back to A, since that forms a loop. By telling C that D is unreachable, A simply guards against the possibility that C might get confused and believe that there is a route through A. This is obvious for a point to point line. But consider the possibility that A and C are connected by a broadcast network such as an Ethernet, and there are other routers on that network. If A has a route through C, it should indicate that D is unreachable when talking to any other router on that network. The other routers on the network can get to C themselves. They would never need to get to C via A. If A's best route is really through C, no other router on that network needs to know that A can reach D. This is fortunate, because it means that the same update message that

is used for C can be used for all other routers on the same network. Thus, update messages can be sent by broadcast.

In general, split horizon with poisoned reverse is safer than simple split horizon. If two routers have routes pointing at each other, advertising reverse routes with a metric of 16 will break the loop immediately. If the reverse routes are simply not advertised, the erroneous routes will have to be eliminated by waiting for a timeout. However, poisoned reverse does have a disadvantage: it increases the size of the routing messages. Consider the case of a campus backbone connecting a number of different buildings. In each building, there is a router connecting the backbone to a local network. Consider what routing updates those routers should broadcast on the backbone network. All that the rest of the network really needs to know about each router is what local networks it is connected to. Using simple split horizon, only those routes would appear in update messages sent by the router to the backbone network. If split horizon with poisoned reverse is used, the router must mention all routes that it learns from the backbone, with metrics of 16. If the system is large, this can result in a large update message, almost all of whose entries indicate unreachable networks.

In a static sense, advertising reverse routes with a metric of 16 provides no additional information. If there are many routers on one broadcast network, these extra entries can use significant bandwidth. The reason they are there is to improve dynamic behavior. When topology changes, mentioning routes that should not go through the router as well as those that should can speed up convergence. However, in some situations, network managers may prefer to accept somewhat slower convergence in order to minimize routing overhead. Thus implementors may at their option implement simple split horizon rather than split horizon with poisoned reverse, or they may provide a configuration option that allows the network manager to choose which behavior to use. It is also permissible to implement hybrid schemes that advertise some reverse routes with a metric of 16 and omit others. An example of such a scheme would be to use a metric of 16 for reverse routes for a certain period of time after routing changes involving them, and thereafter omitting them from updates.

The router requirements RFC [11] specifies that all implementation of RIP must use split horizon and should also use split horizon with poisoned reverse, although there may be a knob to disable poisoned reverse.

3.4.4 Triggered updates

Split horizon with poisoned reverse will prevent any routing loops that involve only two routers. However, it is still possible to end up with patterns in which three routers are engaged in mutual deception. For example, A may believe it has a route through B, B through C, and C through A. Split horizon cannot stop such a loop. This loop will only be resolved when the metric reaches infinity and the network involved is then declared unreachable. Triggered updates are an attempt to speed up this convergence. To get triggered updates, we simply add a rule that whenever a router changes the metric for a route, it is required to send update messages almost immediately, even if it is not yet time for one of the regular update message. (The timing details will differ from protocol to protocol. Some distance vector protocols, including RIP, specify a small time delay, in order to avoid having triggered updates generate excessive network traffic.) Note how this combines with the rules for computing new metrics. Suppose a router's route to destination N goes through router G. If an update arrives from G itself, the receiving router is required to believe the new information, whether the new metric is higher or lower than the old one. If the result is a change in metric, then the receiving router will send triggered updates to all the hosts and routers directly connected to it. They in turn may each send updates to their neighbors. The result is a cascade of triggered updates. It is easy to show which routers and hosts are involved in the cascade. Suppose a router G times out a route to destination N. G will send triggered updates to all of its neighbors. However, the only neighbors who will believe the new information are those whose routes for N go through G. The other routers and hosts will see this as information about a new route that is worse than the one they are already using, and ignore it. The neighbors whose routes go through G will update their metrics and send triggered updates to all of their neighbors. Again, only those neighbors whose routes go through them will pay attention. Thus, the triggered updates will propagate backwards along all paths leading to router G, updating the metrics to infinity. This propagation will stop as soon as it reaches a portion of the network whose route to destination N takes some other path.

If the system could be made to sit still while the cascade of triggered updates happens, it would be possible to prove that counting to infinity will never happen. Bad routes would always be removed immediately, and so no routing loops could form.

Unfortunately, things are not so nice. While the triggered updates are being sent, regular updates may be happening at the same time. Routers that haven't received the triggered update yet will still be sending out information based on the route that no longer exists. It

is possible that after the triggered update has gone through a router, it might receive a normal update from one of these routers that hasn't yet gotten the word. This could reestablish an orphaned remnant of the faulty route. If triggered updates happen quickly enough, this is very unlikely. However, counting to infinity is still possible.

The router requirements RFC [11] specifies that all implementation of RIP must implement triggered update for deleted routes and may implement triggered updates for new routes or change of routes. RIP implementations must also limit the rate which of triggered updates may be transmitted. (see section 3.10.1)

3.5 Protocol Specification

RIP is intended to allow routers to exchange information for computing routes through an IPv4-based network. Any router that uses RIP is assumed to have interfaces to one or more networks, otherwise it isn't really a router. These are referred to as its directly-connected networks. The protocol relies on access to certain information about each of these networks, the most important of which is its metric. The RIP metric of a network is an integer between 1 and 15, inclusive. It is set in some manner not specified in this protocol; however, given the maximum path limit of 15, a value of 1 is usually used. Implementations should allow the system administrator to set the metric of each network. In addition to the metric, each network will have an IPv4 destination address and subnet mask associated with it. These are to be set by the system administrator in a manner not specified in this protocol.

Any host that uses RIP is assumed to have interfaces to one or more networks. These are referred to as its "directly-connected networks". The protocol relies on access to certain information about each of these networks. The most important is its metric or "cost". The metric of a network is an integer between 1 and 15 inclusive. It is set in some manner not specified in this protocol. Most existing implementations always use a metric of 1. New implementations should allow the system administrator to set the cost of each network. In addition to the cost, each network will have an IPv4 network number and a subnet mask associated with it. These are to be set by the system administrator in a manner not specified in this protocol.

Note that the rules specified in section 3.7 assume that there is a single subnet mask applying to each IPv4 network, and that only the subnet masks for directly-connected networks are known. There may be systems that use different subnet masks for different subnets within a single network. There may also be instances where it is desirable

for a system to know the subnets masks of distant networks. Network-wide distribution of routing information which contains different subnet masks is permitted if all routers in the network are running the extensions presented in this document. However, if all routers in the network are not running these extensions distribution of routing information containing different subnet masks must be limited to avoid interoperability problems. See sections 3.7 and 4.3 for the rules governing subnet distribution.

Each router that implements RIP is assumed to have a routing table. This table has one entry for every destination that is reachable throughout the system operating RIP. Each entry contains at least the following information:

- The IPv4 address of the destination.
- A metric, which represents the total cost of getting a datagram from the router to that destination. This metric is the sum of the costs associated with the networks that would be traversed to get to the destination.
- The IPv4 address of the next router along the path to the destination (i.e., the next hop). If the destination is on one of the directly-connected networks, this item is not needed.
- A flag to indicate that information about the route has changed recently. This will be referred to as the "route change flag."
- Various timers associated with the route. See section 3.6 for more details on timers.

The entries for the directly-connected networks are set up by the router using information gathered by means not specified in this protocol. The metric for a directly-connected network is set to the cost of that network. As mentioned, 1 is the usual cost. In that case, the RIP metric reduces to a simple hop-count. More complex metrics may be used when it is desirable to show preference for some networks over others (e.g., to indicate of differences in bandwidth or reliability).

To support the extensions detailed in this document, each entry must additionally contain a subnet mask. The subnet mask allows the router (along with the IPv4 address of the destination) to identify the different subnets within a single network as well as the subnets masks of distant networks.

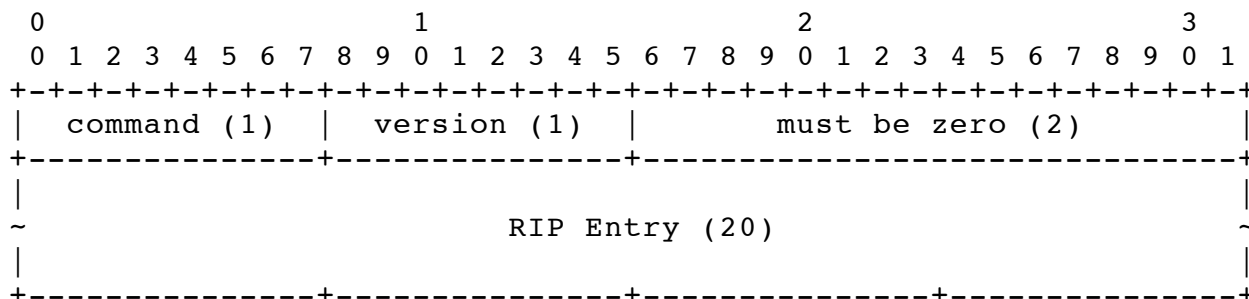
Implementors may also choose to allow the system administrator to enter additional routes. These would most likely be routes to hosts or networks outside the scope of the routing system. They are referred to as "static routes." Entries for destinations other than these initial ones are added and updated by the algorithms described in the following sections.

In order for the protocol to provide complete information on routing, every router in the AS must participate in the protocol. In cases where multiple IGPs are in use, there must be at least one router which can leak routing information between the protocols.

3.6 Message Format

RIP is a UDP-based protocol. Each router that uses RIP has a routing process that sends and receives datagrams on UDP port number 520, the RIP-1/RIP-2 port. All communications intended for another routers's RIP process are sent to the RIP port. All routing update messages are sent from the RIP port. Unsolicited routing update messages have both the source and destination port equal to the RIP port. Update messages sent in response to a request are sent to the port from which the request came. Specific queries may be sent from ports other than the RIP port, but they must be directed to the RIP port on the target machine.

The RIP packet format is:



There may be between 1 and 25 (inclusive) RIP entries. A RIP-1 entry has the following format:

0										1										2										3									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1								
+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-	+	-								
address family identifier (2)										must be zero (2)																													
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																																							
										IPv4 address (4)																													
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																																							
										must be zero (4)																													
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																																							
										must be zero (4)																													
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																																							
										metric (4)																													
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+																																							

Field sizes are given in octets. Unless otherwise specified, fields contain binary integers, in network byte order, with the most-significant octet first (big-endian). Each tick mark represents one bit.

Every message contains a RIP header which consists of a command and a version number. This section of the document describes version 1 of the protocol; section 4 describes the version 2 extensions. The command field is used to specify the purpose of this message. The commands implemented in version 1 and 2 are:

- 1 - request A request for the responding system to send all or part of its routing table.
- 2 - response A message containing all or part of the sender's routing table. This message may be sent in response to a request, or it may be an unsolicited routing update generated by the sender.

For each of these message types, in version 1, the remainder of the datagram contains a list of Route Entries (RTEs). Each RTE in this list contains an Address Family Identifier (AFI), destination IPv4 address, and the cost to reach that destination (metric).

The AFI is the type of address. For RIP-1, only AF_INET (2) is generally supported.

The metric field contains a value between 1 and 15 (inclusive) which specifies the current metric for the destination; or the value 16 (infinity), which indicates that the destination is not reachable.

3.7 Addressing Considerations

Distance vector routing can be used to describe routes to individual hosts or to networks. The RIP protocol allows either of these possibilities. The destinations appearing in request and response messages can be networks, hosts, or a special code used to indicate a default address. In general, the kinds of routes actually used will depend upon the routing strategy used for the particular network. Many networks are set up so that routing information for individual hosts is not needed. If every node on a given network or subnet is accessible through the same routers, then there is no reason to mention individual hosts in the routing tables. However, networks that include point-to-point lines sometimes require routers to keep track of routes to certain nodes. Whether this feature is required depends upon the addressing and routing approach used in the system. Thus, some implementations may choose not to support host routes. If host routes are not supported, they are to be dropped when they are received in response messages (see section 3.7.2).

The RIP-1 packet format does not distinguish among various types of address. Fields that are labeled "address" can contain any of the following:

host address subnet number network number zero (default route)

Entities which use RIP-1 are assumed to use the most specific information available when routing a datagram. That is, when routing a datagram, its destination address must first be checked against the list of node addresses. Then it must be checked to see whether it matches any known subnet or network number. Finally, if none of these match, the default route is used.

When a node evaluates information that it receives via RIP-1, its interpretation of an address depends upon whether it knows the subnet mask that applies to the net. If so, then it is possible to determine the meaning of the address. For example, consider net 128.6. It has a subnet mask of 255.255.255.0. Thus 128.6.0.0 is a network number, 128.6.4.0 is a subnet number, and 128.6.4.1 is a node address. However, if the node does not know the subnet mask, evaluation of an address may be ambiguous. If there is a non-zero node part, there is no clear way to determine whether the address represents a subnet number or a node address. As a subnet number would be useless without the subnet mask, addresses are assumed to represent nodes in this situation. In order to avoid this sort of ambiguity, when using version 1, nodes must not send subnet routes to nodes that cannot be expected to know the appropriate subnet mask. Normally hosts only know the subnet masks for directly-connected networks. Therefore, unless special provisions have been made,

routes to a subnet must not be sent outside the network of which the subnet is a part. RIP-2 (see section 4) eliminates the subnet/host ambiguity by including the subnet mask in the routing entry.

This "subnet filtering" is carried out by the routers at the "border" of the subnetted network. These are routers which connect that network with some other network. Within the subnetted network, each subnet is treated as an individual network. Routing entries for each subnet are circulated by RIP. However, border routers send only a single entry for the network as a whole to nodes in other networks. This means that a border router will send different information to different neighbors. For neighbors connected to the subnetted network, it generates a list of all subnets to which it is directly connected, using the subnet number. For neighbors connected to other networks, it makes a single entry for the network as a whole, showing the metric associated with that network. This metric would normally be the smallest metric for the subnets to which the router is attached.

Similarly, border routers must not mention host routes for nodes within one of the directly-connected networks in messages to other networks. Those routes will be subsumed by the single entry for the network as a whole.

The router requirements RFC [11] specifies that all implementation of RIP should support host routes but if they do not then they must ignore any received host routes.

The special address 0.0.0.0 is used to describe a default route. A default route is used when it is not convenient to list every possible network in the RIP updates, and when one or more closely-connected routers in the system are prepared to handle traffic to the networks that are not listed explicitly. These routers should create RIP entries for the address 0.0.0.0, just as if it were a network to which they are connected. The decision as to how routers create entries for 0.0.0.0 is left to the implementor. Most commonly, the system administrator will be provided with a way to specify which routers should create entries for 0.0.0.0; however, other mechanisms are possible. For example, an implementor might decide that any router which speaks BGP should be declared to be a default router. It may be useful to allow the network administrator to choose the metric to be used in these entries. If there is more than one default router, this will make it possible to express a preference for one over the other. The entries for 0.0.0.0 are handled by RIP in exactly the same manner as if there were an actual network with this address. System administrators should take care to make sure that routes to 0.0.0.0 do not propagate further than is intended. Generally, each autonomous system has its own preferred default

router. Thus, routes involving 0.0.0.0 should generally not leave the boundary of an autonomous system. The mechanisms for enforcing this are not specified in this document.

3.8 Timers

This section describes all events that are triggered by timers.

Every 30 seconds, the RIP process is awakened to send an unsolicited Response message containing the complete routing table (see section 3.9 on Split Horizon) to every neighboring router. When there are many routers on a single network, there is a tendency for them to synchronize with each other such that they all issue updates at the same time. This can happen whenever the 30 second timer is affected by the processing load on the system. It is undesirable for the update messages to become synchronized, since it can lead to unnecessary collisions on broadcast networks. Therefore, implementations are required to take one of two precautions:

- The 30-second updates are triggered by a clock whose rate is not affected by system load or the time required to service the previous update timer.
- The 30-second timer is offset by a small random time (+/- 0 to 5 seconds) each time it is set. (Implementors may wish to consider even larger variation in the light of recent research results [10])

There are two timers associated with each route, a "timeout" and a "garbage-collection" time. Upon expiration of the timeout, the route is no longer valid; however, it is retained in the routing table for a short time so that neighbors can be notified that the route has been dropped. Upon expiration of the garbage-collection timer, the route is finally removed from the routing table.

The timeout is initialized when a route is established, and any time an update message is received for the route. If 180 seconds elapse from the last time the timeout was initialized, the route is considered to have expired, and the deletion process described below begins for that route.

Deletions can occur for one of two reasons: the timeout expires, or the metric is set to 16 because of an update received from the current router (see section 3.7.2 for a discussion of processing updates from other routers). In either case, the following events happen:

- The garbage-collection timer is set for 120 seconds.
- The metric for the route is set to 16 (infinity). This causes the route to be removed from service.
- The route change flag is set to indicate that this entry has been changed.
- The output process is signalled to trigger a response.

Until the garbage-collection timer expires, the route is included in all updates sent by this router. When the garbage-collection timer expires, the route is deleted from the routing table.

Should a new route to this network be established while the garbage-collection timer is running, the new route will replace the one that is about to be deleted. In this case the garbage-collection timer must be cleared.

Triggered updates also use a small timer; however, this is best described in section 3.9.1.

3.9 Input Processing

This section will describe the handling of datagrams received on the RIP port. Processing will depend upon the value in the command field.

See sections 4.6 and 5.1 for details on handling version numbers.

3.9.1 Request Messages

A Request is used to ask for a response containing all or part of a router's routing table. Normally, Requests are sent as broadcasts (multicasts for RIP-2), from the RIP port, by routers which have just come up and are seeking to fill in their routing tables as quickly as possible. However, there may be situations (e.g., router monitoring) where the routing table of only a single router is needed. In this case, the Request should be sent directly to that router from a UDP port other than the RIP port. If such a Request is received, the router responds directly to the requestor's address and port.

The Request is processed entry by entry. If there are no entries, no response is given. There is one special case. If there is exactly one entry in the request, and it has an address family identifier of zero and a metric of infinity (i.e., 16), then this is a request to send the entire routing table. In that case, a call is made to the output process to send the routing table to the requesting

address/port. Except for this special case, processing is quite simple. Examine the list of RTEs in the Request one by one. For each entry, look up the destination in the router's routing database and, if there is a route, put that route's metric in the metric field of the RTE. If there is no explicit route to the specified destination, put infinity in the metric field. Once all the entries have been filled in, change the command from Request to Response and send the datagram back to the requestor.

Note that there is a difference in metric handling for specific and whole-table requests. If the request is for a complete routing table, normal output processing is done, including Split Horizon (see section 3.9 on Split Horizon). If the request is for specific entries, they are looked up in the routing table and the information is returned as is; no Split Horizon processing is done. The reason for this distinction is the expectation that these requests are likely to be used for different purposes. When a router first comes up, it multicasts a Request on every connected network asking for a complete routing table. It is assumed that these complete routing tables are to be used to update the requestor's routing table. For this reason, Split Horizon must be done. It is further assumed that a Request for specific networks is made only by diagnostic software, and is not used for routing. In this case, the requester would want to know the exact contents of the routing table and would not want any information hidden or modified.

3.9.2 Response Messages

A Response can be received for one of several different reasons:

- response to a specific query
- regular update (unsolicited response)
- triggered update caused by a route change

Processing is the same no matter why the Response was generated.

Because processing of a Response may update the router's routing table, the Response must be checked carefully for validity. The Response must be ignored if it is not from the RIP port. The datagram's IPv4 source address should be checked to see whether the datagram is from a valid neighbor; the source of the datagram must be on a directly-connected network. It is also worth checking to see whether the response is from one of the router's own addresses. Interfaces on broadcast networks may receive copies of their own broadcasts/multicasts immediately. If a router processes its own output as new input, confusion is likely so such datagrams must be ignored.

Once the datagram as a whole has been validated, process the RTEs in the Response one by one. Again, start by doing validation. Incorrect metrics and other format errors usually indicate misbehaving neighbors and should probably be brought to the administrator's attention. For example, if the metric is greater than infinity, ignore the entry but log the event. The basic validation tests are:

- is the destination address valid (e.g., unicast; not net 0 or 127)
- is the metric valid (i.e., between 1 and 16, inclusive)

If any check fails, ignore that entry and proceed to the next. Again, logging the error is probably a good idea.

Once the entry has been validated, update the metric by adding the cost of the network on which the message arrived. If the result is greater than infinity, use infinity. That is,

`metric = MIN (metric + cost, infinity)`

Now, check to see whether there is already an explicit route for the destination address. If there is no such route, add this route to the routing table, unless the metric is infinity (there is no point in adding a route which is unusable). Adding a route to the routing table consists of:

- Setting the destination address to the destination address in the RTE
- Setting the metric to the newly calculated metric (as described above)
- Set the next hop address to be the address of the router from which the datagram came
- Initialize the timeout for the route. If the garbage-collection timer is running for this route, stop it (see section 3.6 for a discussion of the timers)
- Set the route change flag
- Signal the output process to trigger an update (see section 3.8.1)

If there is an existing route, compare the next hop address to the address of the router from which the datagram came. If this datagram is from the same router as the existing route, reinitialize the timeout. Next, compare the metrics. If the datagram is from the same router as the existing route, and the new metric is different

than the old one; or, if the new metric is lower than the old one; do the following actions:

- Adopt the route from the datagram (i.e., put the new metric in and adjust the next hop address, if necessary).
- Set the route change flag and signal the output process to trigger an update
- If the new metric is infinity, start the deletion process (described above); otherwise, re-initialize the timeout

If the new metric is infinity, the deletion process begins for the route, which is no longer used for routing packets. Note that the deletion process is started only when the metric is first set to infinity. If the metric was already infinity, then a new deletion process is not started.

If the new metric is the same as the old one, it is simplest to do nothing further (beyond re-initializing the timeout, as specified above); but, there is a heuristic which could be applied. Normally, it is senseless to replace a route if the new route has the same metric as the existing route; this would cause the route to bounce back and forth, which would generate an intolerable number of triggered updates. However, if the existing route is showing signs of timing out, it may be better to switch to an equally-good alternative route immediately, rather than waiting for the timeout to happen. Therefore, if the new metric is the same as the old one, examine the timeout for the existing route. If it is at least halfway to the expiration point, switch to the new route. This heuristic is optional, but highly recommended.

Any entry that fails these tests is ignored, as it is no better than the current route.

3.10 Output Processing

This section describes the processing used to create response messages that contain all or part of the routing table. This processing may be triggered in any of the following ways:

- By input processing, when a Request is received (this Response is unicast to the requestor; see section 3.7.1)
- By the regular routing update (broadcast/multicast every 30 seconds) router.
- By triggered updates (broadcast/multicast when a route changes)

When a Response is to be sent to all neighbors (i.e., a regular or triggered update), a Response message is directed to the router at the far end of each connected point-to-point link, and is broadcast (multicast for RIP-2) on all connected networks which support broadcasting. Thus, one Response is prepared for each directly-connected network, and sent to the appropriate address (direct or broadcast/multicast). In most cases, this reaches all neighboring routers. However, there are some cases where this may not be good enough. This may involve a network that is not a broadcast network (e.g., the ARPANET), or a situation involving dumb routers. In such cases, it may be necessary to specify an actual list of neighboring routers and send a datagram to each one explicitly. It is left to the implementor to determine whether such a mechanism is needed, and to define how the list is specified.

3.10.1 Triggered Updates

Triggered updates require special handling for two reasons. First, experience shows that triggered updates can cause excessive load on networks with limited capacity or networks with many routers on them. Therefore, the protocol requires that implementors include provisions to limit the frequency of triggered updates. After a triggered update is sent, a timer should be set for a random interval between 1 and 5 seconds. If other changes that would trigger updates occur before the timer expires, a single update is triggered when the timer expires. The timer is then reset to another random value between 1 and 5 seconds. A triggered update should be suppressed if a regular update is due by the time the triggered update would be sent.

Second, triggered updates do not need to include the entire routing table. In principle, only those routes which have changed need to be included. Therefore, messages generated as part of a triggered update must include at least those routes that have their route change flag set. They may include additional routes, at the discretion of the implementor; however, sending complete routing updates is strongly discouraged. When a triggered update is processed, messages should be generated for every directly-connected network. Split Horizon processing is done when generating triggered updates as well as normal updates (see section 3.9). If, after Split Horizon processing for a given network, a changed route will appear unchanged on that network (e.g., it appears with an infinite metric), the route need not be sent. If no routes need be sent on that network, the update may be omitted. Once all of the triggered updates have been generated, the route change flags should be cleared.

If input processing is allowed while output is being generated, appropriate interlocking must be done. The route change flags should not be changed as a result of processing input while a triggered update message is being generated.

The only difference between a triggered update and other update messages is the possible omission of routes that have not changed. The remaining mechanisms, described in the next section, must be applied to all updates.

3.10.2 Generating Response Messages

This section describes how a Response message is generated for a particular directly-connected network:

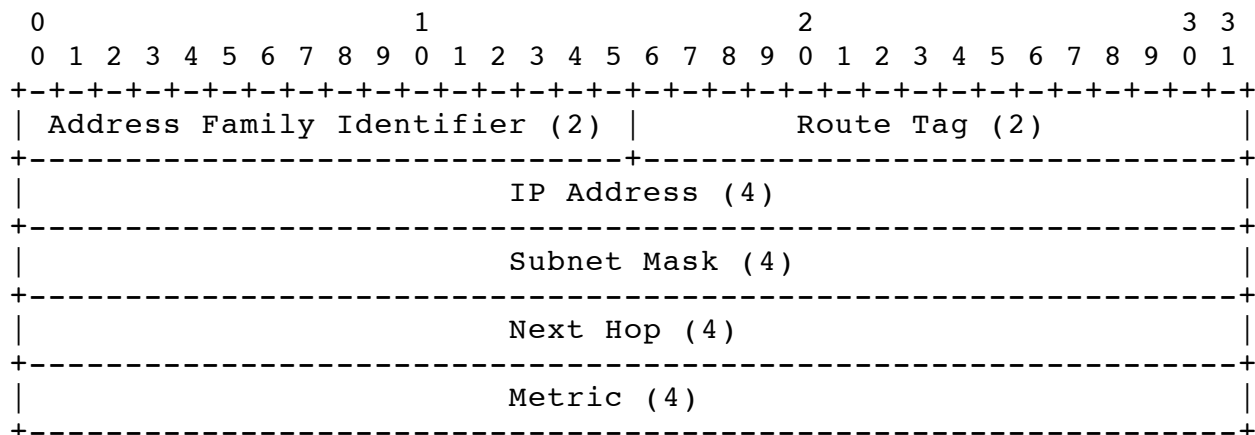
Set the version number to either 1 or 2. The mechanism for deciding which version to send is implementation specific; however, if this is the Response to a Request, the Response version should match the Request version. Set the command to Response. Set the bytes labeled "must be zero" to zero. Start filling in RTEs. Recall that there is a limit of 25 RTEs to a Response; if there are more, send the current Response and start a new one. There is no defined limit to the number of datagrams which make up a Response.

To fill in the RTEs, examine each route in the routing table. If a triggered update is being generated, only entries whose route change flags are set need be included. If, after Split Horizon processing, the route should not be included, skip it. If the route is to be included, then the destination address and metric are put into the RTE. Routes must be included in the datagram even if their metrics are infinite.

4. Protocol Extensions

This section does not change the RIP protocol per se. Rather, it provides extensions to the message format which allows routers to share important additional information.

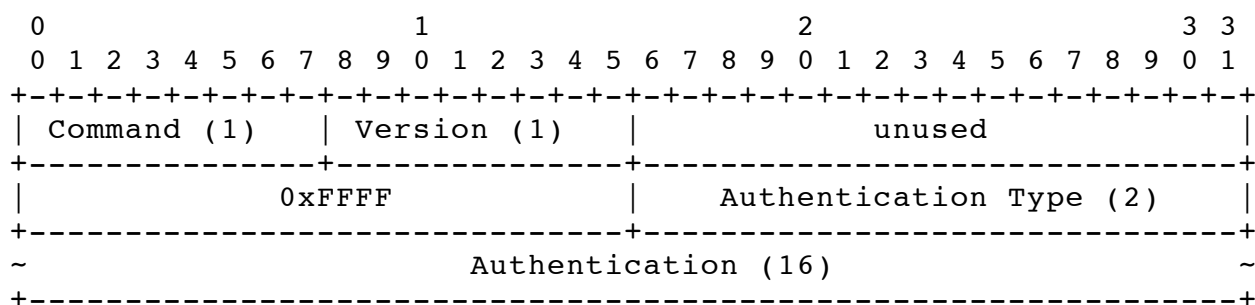
The same header format is used for RIP-1 and RIP-2 messages (see section 3.4). The format for the 20-octet route entry (RTE) for RIP-2 is:



The Address Family Identifier, IP Address, and Metric all have the meanings defined in section 3.4. The Version field will specify version number 2 for RIP messages which use authentication or carry information in any of the newly defined fields.

4.1 Authentication

Since authentication is a per message function, and since there is only one 2-octet field available in the message header, and since any reasonable authentication scheme will require more than two octets, the authentication scheme for RIP version 2 will use the space of an entire RIP entry. If the Address Family Identifier of the first (and only the first) entry in the message is 0xFFFF, then the remainder of the entry contains the authentication. This means that there can be, at most, 24 RIP entries in the remainder of the message. If authentication is not in use, then no entries in the message should have an Address Family Identifier of 0xFFFF. A RIP message which contains an authentication entry would begin with the following format:



Currently, the only Authentication Type is simple password and it is type 2. The remaining 16 octets contain the plain text password. If the password is under 16 octets, it must be left-justified and padded to the right with nulls (0x00).

4.2 Route Tag

The Route Tag (RT) field is an attribute assigned to a route which must be preserved and readadvertised with a route. The intended use of the Route Tag is to provide a method of separating "internal" RIP routes (routes for networks within the RIP routing domain) from "external" RIP routes, which may have been imported from an EGP or another IGP.

Routers supporting protocols other than RIP should be configurable to allow the Route Tag to be configured for routes imported from different sources. For example, routes imported from EGP or BGP should be able to have their Route Tag either set to an arbitrary value, or at least to the number of the Autonomous System from which the routes were learned.

Other uses of the Route Tag are valid, as long as all routers in the RIP domain use it consistently. This allows for the possibility of a BGP-RIP protocol interactions document, which would describe methods for synchronizing routing in a transit network.

4.3 Subnet mask

The Subnet Mask field contains the subnet mask which is applied to the IP address to yield the non-host portion of the address. If this field is zero, then no subnet mask has been included for this entry.

On an interface where a RIP-1 router may hear and operate on the information in a RIP-2 routing entry the following rules apply:

- 1) information internal to one network must never be advertised into another network,

- 2) information about a more specific subnet may not be advertised where RIP-1 routers would consider it a host route, and
- 3) supernet routes (routes with a netmask less specific than the "natural" network mask) must not be advertised where they could be misinterpreted by RIP-1 routers.

4.4 Next Hop

The immediate next hop IP address to which packets to the destination specified by this route entry should be forwarded. Specifying a value of 0.0.0.0 in this field indicates that routing should be via the originator of the RIP advertisement. An address specified as a next hop must, per force, be directly reachable on the logical subnet over which the advertisement is made.

The purpose of the Next Hop field is to eliminate packets being routed through extra hops in the system. It is particularly useful when RIP is not being run on all of the routers on a network. A simple example is given in Appendix A. Note that Next Hop is an "advisory" field. That is, if the provided information is ignored, a possibly sub-optimal, but absolutely valid, route may be taken. If the received Next Hop is not directly reachable, it should be treated as 0.0.0.0.

4.5 Multicasting

In order to reduce unnecessary load on those hosts which are not listening to RIP-2 messages, an IP multicast address will be used for periodic broadcasts. The IP multicast address is 224.0.0.9. Note that IGMP is not needed since these are inter-router messages which are not forwarded.

On NBMA networks, unicast addressing may be used. However, if a response addressed to the RIP-2 multicast address is received, it should be accepted.

In order to maintain backwards compatibility, the use of the multicast address will be configurable, as described in section 5.1. If multicasting is used, it should be used on all interfaces which support it.

4.6 Queries

If a RIP-2 router receives a RIP-1 Request, it should respond with a RIP-1 Response. If the router is configured to send only RIP-2 messages, it should not respond to a RIP-1 Request.

5. Compatibility

RFC [1] showed considerable forethought in its specification of the handling of version numbers. It specifies that RIP messages of version 0 are to be discarded, that RIP messages of version 1 are to be discarded if any Must Be Zero (MBZ) field is non-zero, and that RIP messages of any version greater than 1 should not be discarded simply because an MBZ field contains a value other than zero. This means that the new version of RIP is totally backwards compatible with existing RIP implementations which adhere to this part of the specification.

5.1 Compatibility Switch

A compatibility switch is necessary for two reasons. First, there are implementations of RIP-1 in the field which do not follow RFC [1] as described above. Second, the use of multicasting would prevent RIP-1 systems from receiving RIP-2 updates (which may be a desired feature in some cases). This switch should be configurable on a per-interface basis.

The switch has four settings: RIP-1, in which only RIP-1 messages are sent; RIP-1 compatibility, in which RIP-2 messages are broadcast; RIP-2, in which RIP-2 messages are multicast; and "none", which disables the sending of RIP messages. It is recommended that the default setting be either RIP-1 or RIP-2, but not RIP-1 compatibility. This is because of the potential problems which can occur on some topologies. RIP-1 compatibility should only be used when all of the consequences of its use are well understood by the network administrator.

For completeness, routers should also implement a receive control switch which would determine whether to accept, RIP-1 only, RIP-2 only, both, or none. It should also be configurable on a per-interface basis. It is recommended that the default be compatible with the default chosen for sending updates.

5.2 Authentication

The following algorithm should be used to authenticate a RIP message. If the router is not configured to authenticate RIP-2 messages, then RIP-1 and unauthenticated RIP-2 messages will be accepted; authenticated RIP-2 messages shall be discarded. If the router is configured to authenticate RIP-2 messages, then RIP-1 messages and RIP-2 messages which pass authentication testing shall be accepted; unauthenticated and failed authentication RIP-2 messages shall be discarded. For maximum security, RIP-1 messages should be ignored

when authentication is in use (see section 4.1); otherwise, the routing information from authenticated messages will be propagated by RIP-1 routers in an unauthenticated manner.

Since an authentication entry is marked with an Address Family Identifier of 0xFFFF, a RIP-1 system would ignore this entry since it would belong to an address family other than IP. It should be noted, therefore, that use of authentication will not prevent RIP-1 systems from seeing RIP-2 messages. If desired, this may be done using multicasting, as described in sections 4.5 and 5.1.

5.3 Larger Infinity

While on the subject of compatibility, there is one item which people have requested: increasing infinity. The primary reason that this cannot be done is that it would violate backwards compatibility. A larger infinity would obviously confuse older versions of rip. At best, they would ignore the route as they would ignore a metric of 16. There was also a proposal to make the Metric a single octet and reuse the high three octets, but this would break any implementations which treat the metric as a 4-octet entity.

5.4 Addressless Links

As in RIP-1, addressless links will not be supported by RIP-2.

6. Interaction between version 1 and version 2

Because version 1 packets do not contain subnet information, the semantics employed by routers on networks that contain both version 1 and version 2 networks should be limited to that of version 1. Otherwise it is possible either to create blackhole routes (i.e., routes for networks that do not exist) or to create excessive routing information in a version 1 environment.

Some implementations attempt to automatically summarize groups of adjacent routes into single entries, the goal being to reduce the total number of entries. This is called auto-summarization.

Specifically, when using both version 1 and version 2 within a network, a single subnet mask should be used throughout the network. In addition, auto-summarization mechanisms should be disabled for such networks, and implementations must provide mechanisms to disable auto-summarization.

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7. Security Considerations

The basic RIP protocol is not a secure protocol. To bring RIP-2 in line with more modern routing protocols, an extensible authentication mechanism has been incorporated into the protocol enhancements. This mechanism is described in sections 4.1 and 5.2. Security is further enhanced by the mechanism described in [3].

Appendix A

This is a simple example of the use of the next hop field in a rip entry.



Assume that IR1, IR2, and IR3 are all "internal" routers which are under one administration (e.g. a campus) which has elected to use RIP-2 as its IGP. XR1, XR2, and XR3, on the other hand, are under separate administration (e.g. a regional network, of which the campus is a member) and are using some other routing protocol (e.g. OSPF). XR1, XR2, and XR3 exchange routing information among themselves such that they know that the best routes to networks N1 and N2 are via XR1, to N3, N4, and N5 are via XR2, and to N6 and N7 are via XR3. By setting the Next Hop field correctly (to XR2 for N3/N4/N5, to XR3 for N6/N7), only XR1 need exchange RIP-2 routes with IR1/IR2/IR3 for routing to occur without additional hops through XR1. Without the Next Hop (for example, if RIP-1 were used) it would be necessary for XR2 and XR3 to also participate in the RIP-2 protocol to eliminate extra hops.

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